Approximating price series, part 1: least-squares cubic spline (fixed knots) and free-knot cubic spline

Approximation study, 2020-01-02 to 2023-07-31 daily closes, three tickers. All numbers recomputed from the saved fits (approx_study/fits/*.json).

1. What is being measured

This is an in-sample APPROXIMATION task, not forecasting. Each model receives the entire close-price series of a stock, p(t), with t rescaled to [0, 1] (905 points for CBA, 901 for TSLA and LLY; prices in the file's whole-series Min-Max scale, 0 = lowest close of the period, 1 = highest), and must reproduce it as a function of t with a fixed number of parameters. The score is the RMSE and the maximum absolute error between the fitted curve and the actual series, and how both fall as the parameter budget grows (about 15, 30, 60, 120, 240). There is no train/test split and no prediction; in-sample fit says nothing about forecasting. For scale, the dotted line in the error plots is the one-day-ahead naive forecasting error of the same series (0.0201 CBA, 0.0255 TSLA, 0.0148 LLY).

2. The two models in this document

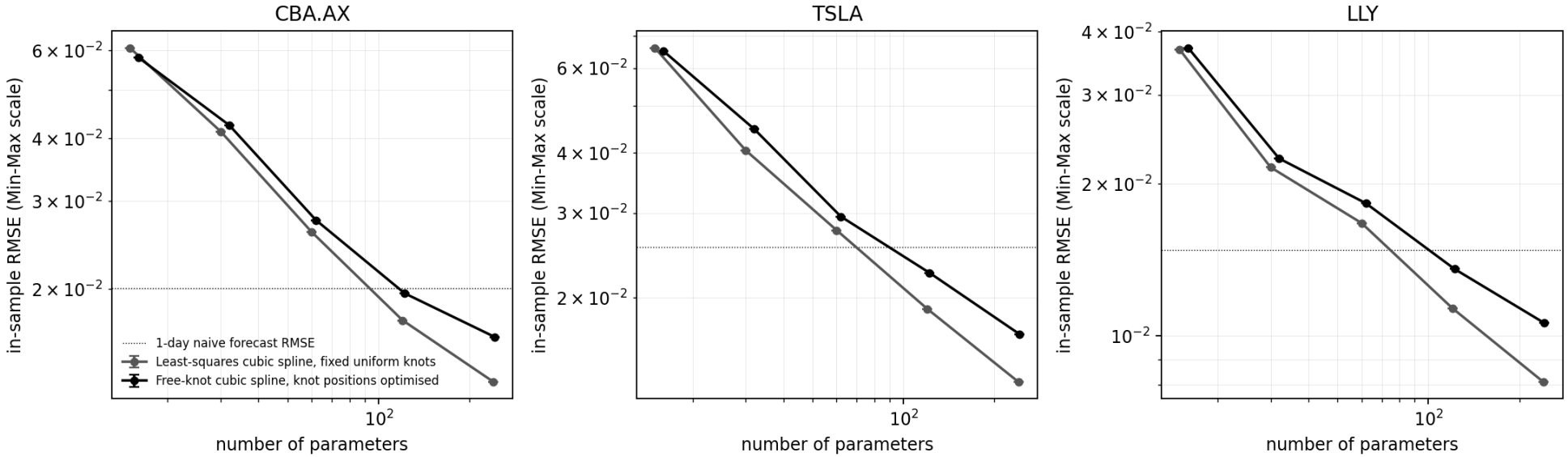
Least-squares cubic spline, fixed uniform knots
Free-knot cubic spline, knot positions optimised

• Both models are classical cubic B-splines fitted by least squares to the whole series p(t), t in [0, 1]. They are deterministic (no seeds).
• Fixed-knot: G equal intervals on [0, 1]; the only free parameters are the G + 3 B-spline coefficients.
• Free-knot: G intervals whose G - 1 interior knot positions are also optimised (12 random starts, L-BFGS-B followed by Nelder-Mead, coefficients re-solved by least squares inside every evaluation; knots kept at least 4 data points apart). Parameters = (G + 3) coefficients + (G - 1) knot positions.
• The one thing that changes between the two models is whether the knots may move. Because a knot position costs a parameter, the free-knot spline has FEWER intervals than the fixed-knot spline at the same budget (e.g. 30 vs 57 intervals at ~60 parameters).

3. Settings per budget

target budget	15	30	60	120	
Least-squares cubic spline, fixed uniform knots actual params	15	30	60	120	
Free-knot cubic spline, knot positions optimised actual params	16	32	62	122	
setting	G (intervals): fixed-knot 12, free-knot 7	G (intervals): fixed-knot 27, free-knot 15	G (intervals): fixed-knot 57, free-knot 30	G (intervals): fixed-knot 117, free-knot 60	

4. Error versus number of parameters (all tickers)



Points are the mean over seeds, bars the sample standard deviation across seeds (absent for deterministic models). Log-log axes: a straight line means error falls as a power of the parameter count.

Slope of log RMSE against log parameters, budgets 30-240 (how fast each model improves with size):

model	CBA.AX	TSLA	LLY	mean
Least-squares cubic spline, fixed uniform knots	-0.56	-0.53	-0.48	-0.52
Free-knot cubic spline, knot positions optimised	-0.48	-0.48	-0.38	-0.45

5. Every fit: full table

One row per fit (ticker, model, budget, seed). RMSE and max |error| in the ticker's Min-Max scale; seconds = wall-clock fit time.

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
CBA.AX	lsq	15	0	15	0.0607	0.2258	0.1	11 interior knots
CBA.AX	lsq	30	0	30	0.0413	0.1724	0.0	26 interior knots
CBA.AX	lsq	60	0	60	0.0260	0.1159	0.0	56 interior knots
CBA.AX	lsq	120	0	120	0.0173	0.0824	0.0	116 interior knots
CBA.AX	lsq	240	0	240	0.0131	0.0634	0.0	236 interior knots
CBA.AX	fks	15	0	16	0.0581	0.1749	5.6	6 interior knots
CBA.AX	fks	30	0	32	0.0425	0.1623	11.5	14 interior knots
CBA.AX	fks	60	0	62	0.0275	0.1053	4.0	29 interior knots
CBA.AX	fks	120	0	122	0.0196	0.0950	3.0	59 interior knots
CBA.AX	fks	240	0	242	0.0161	0.0797	1.6	119 interior knots
TSLA	lsq	15	0	15	0.0661	0.2474	0.0	11 interior knots
TSLA	lsq	30	0	30	0.0405	0.1665	0.0	26 interior knots
TSLA	lsq	60	0	60	0.0276	0.1170	0.0	56 interior knots
TSLA	lsq	120	0	120	0.0189	0.0860	0.1	116 interior knots
TSLA	lsq	240	0	240	0.0134	0.0711	0.0	236 interior knots
TSLA	fks	15	0	16	0.0651	0.2400	5.8	6 interior knots
TSLA	fks	30	0	32	0.0448	0.1592	12.6	14 interior knots
TSLA	fks	60	0	62	0.0295	0.1213	3.7	29 interior knots
TSLA	fks	120	0	122	0.0225	0.1101	3.7	59 interior knots
TSLA	fks	240	0	242	0.0168	0.0764	4.9	119 interior knots
LLY	lsq	15	0	15	0.0369	0.1154	0.0	11 interior knots
LLY	lsq	30	0	30	0.0216	0.0739	0.0	26 interior knots
LLY	lsq	60	0	60	0.0167	0.0646	0.0	56 interior knots
LLY	lsq	120	0	120	0.0114	0.0481	0.0	116 interior knots
LLY	lsq	240	0	240	0.0081	0.0353	0.0	236 interior knots
LLY	fks	15	0	16	0.0372	0.1066	4.5	6 interior knots

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
LLY	fks	30	0	32	0.0225	0.0877	10.7	14 interior knots
LLY	fks	60	0	62	0.0183	0.0689	1.7	29 interior knots
LLY	fks	120	0	122	0.0136	0.0534	1.4	59 interior knots
LLY	fks	240	0	242	0.0106	0.0449	1.5	119 interior knots

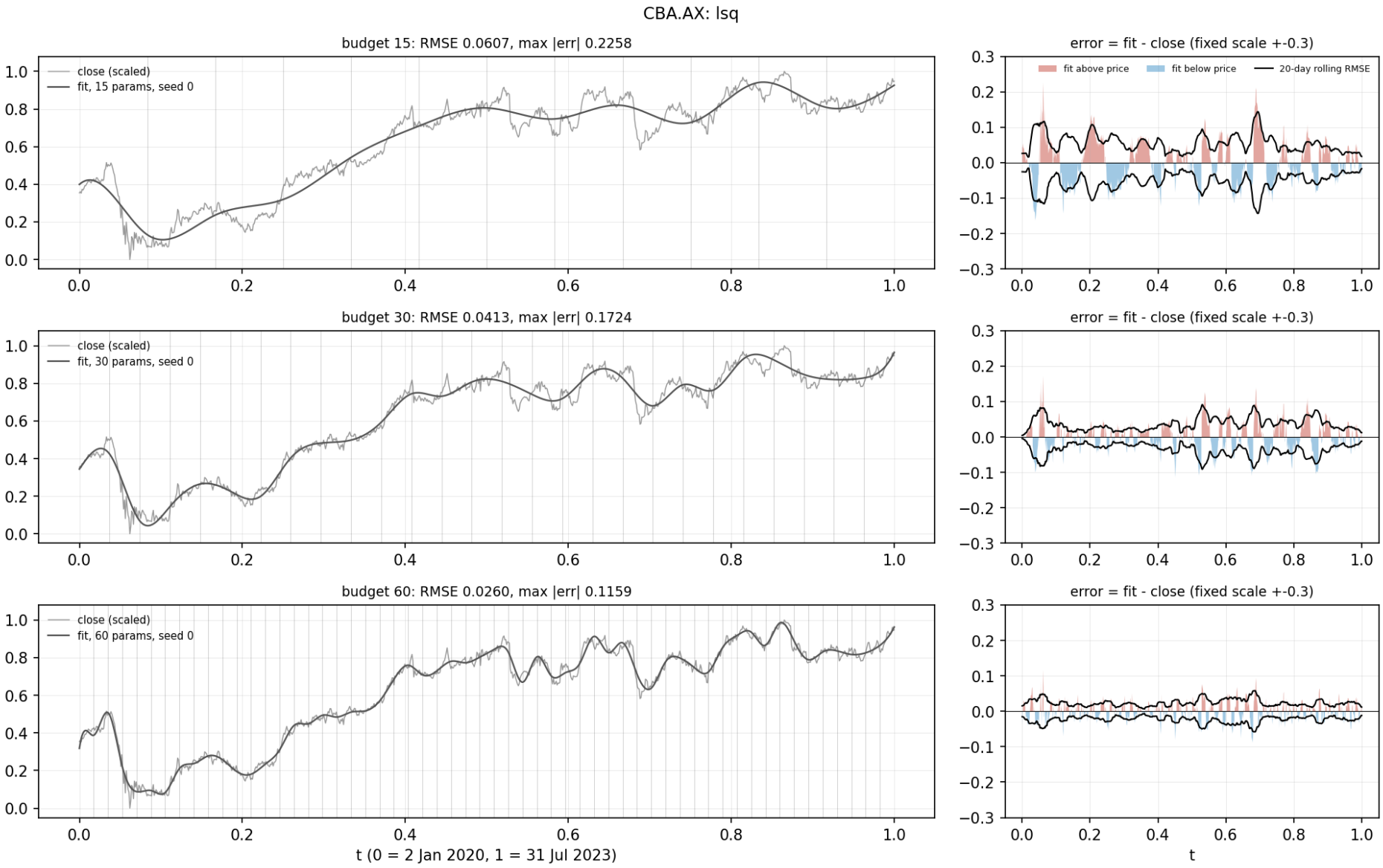
6. Summary per ticker and budget (mean +- sd over seeds)

ticker	budget	Least-squares cubic spline, fixed uniform knots RMSE	max |err|	params	Free-knot cubic spline, knot positions optimised RMSE	max |err|	params
CBA.AX	15	0.0607	0.2258	15	0.0581	0.1749	16
CBA.AX	30	0.0413	0.1724	30	0.0425	0.1623	32
CBA.AX	60	0.0260	0.1159	60	0.0275	0.1053	62
CBA.AX	120	0.0173	0.0824	120	0.0196	0.0950	122
CBA.AX	240	0.0131	0.0634	240	0.0161	0.0797	242
TSLA	15	0.0661	0.2474	15	0.0651	0.2400	16
TSLA	30	0.0405	0.1665	30	0.0448	0.1592	32
TSLA	60	0.0276	0.1170	60	0.0295	0.1213	62
TSLA	120	0.0189	0.0860	120	0.0225	0.1101	122
TSLA	240	0.0134	0.0711	240	0.0168	0.0764	242
LLY	15	0.0369	0.1154	15	0.0372	0.1066	16
LLY	30	0.0216	0.0739	30	0.0225	0.0877	32
LLY	60	0.0167	0.0646	60	0.0183	0.0689	62
LLY	120	0.0114	0.0481	120	0.0136	0.0534	122
LLY	240	0.0081	0.0353	240	0.0106	0.0449	242

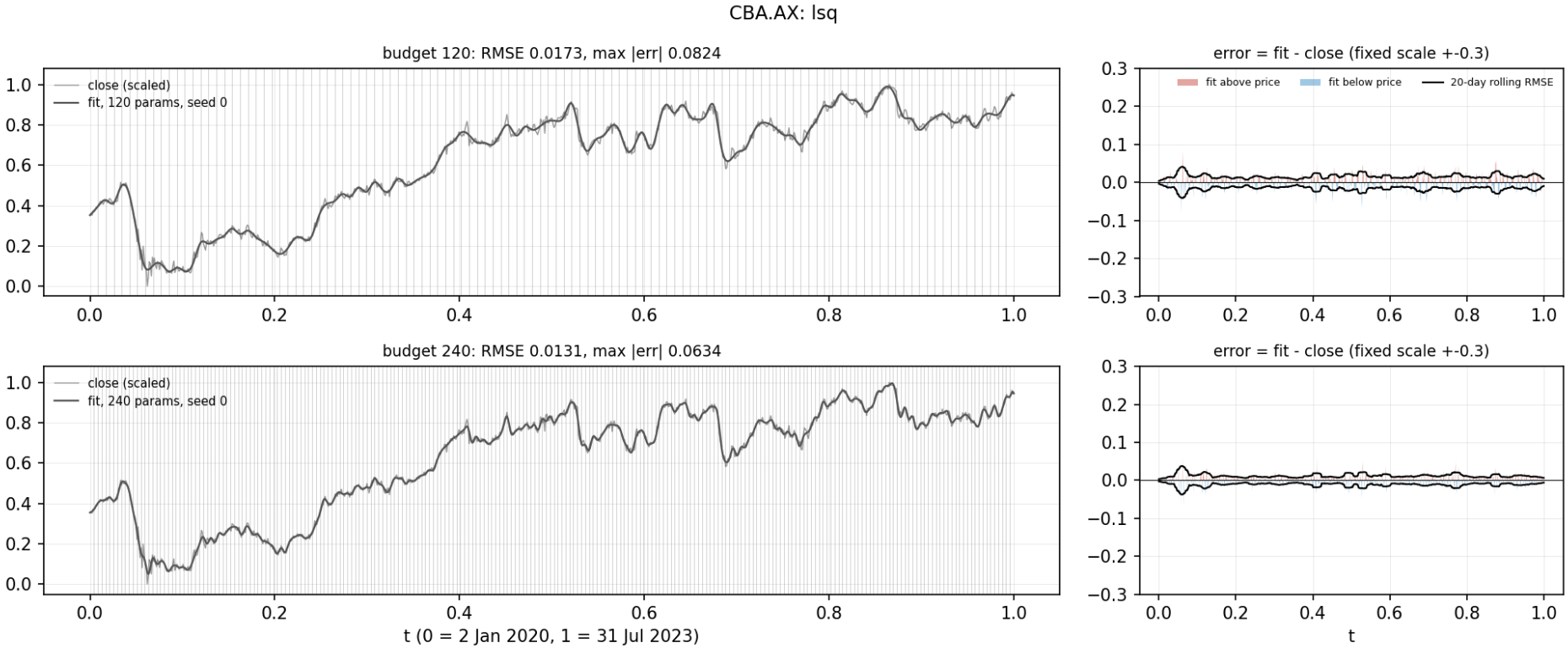
7. The fits themselves (following pages)

For every ticker, model and budget: left, the fitted curve over the actual series (best seed where there are seeds; all seed RMSEs in the title); right, the error fit - close on a fixed +-0.3 scale so panels are comparable: red shading where the fit sits above the price, blue where below, and the black envelope is the 20-day rolling RMSE (+-), i.e. the local size of the error with the day-to-day jitter smoothed out. Vertical lines in spline panels are knot positions.

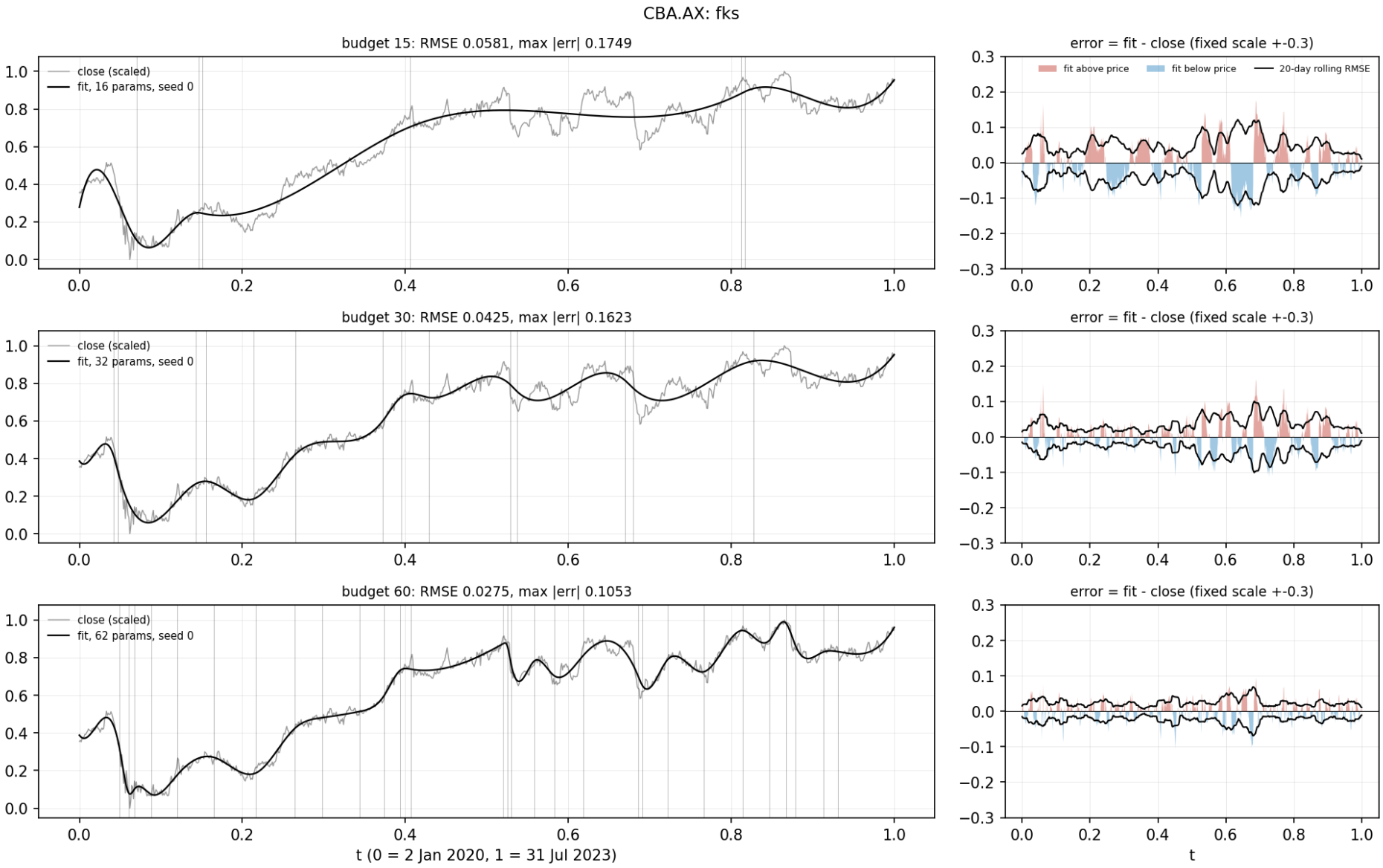
CBA.AX (Commonwealth Bank, benchmark series), Least-squares cubic spline, fixed uniform knots (budgets 15, 30, 60)



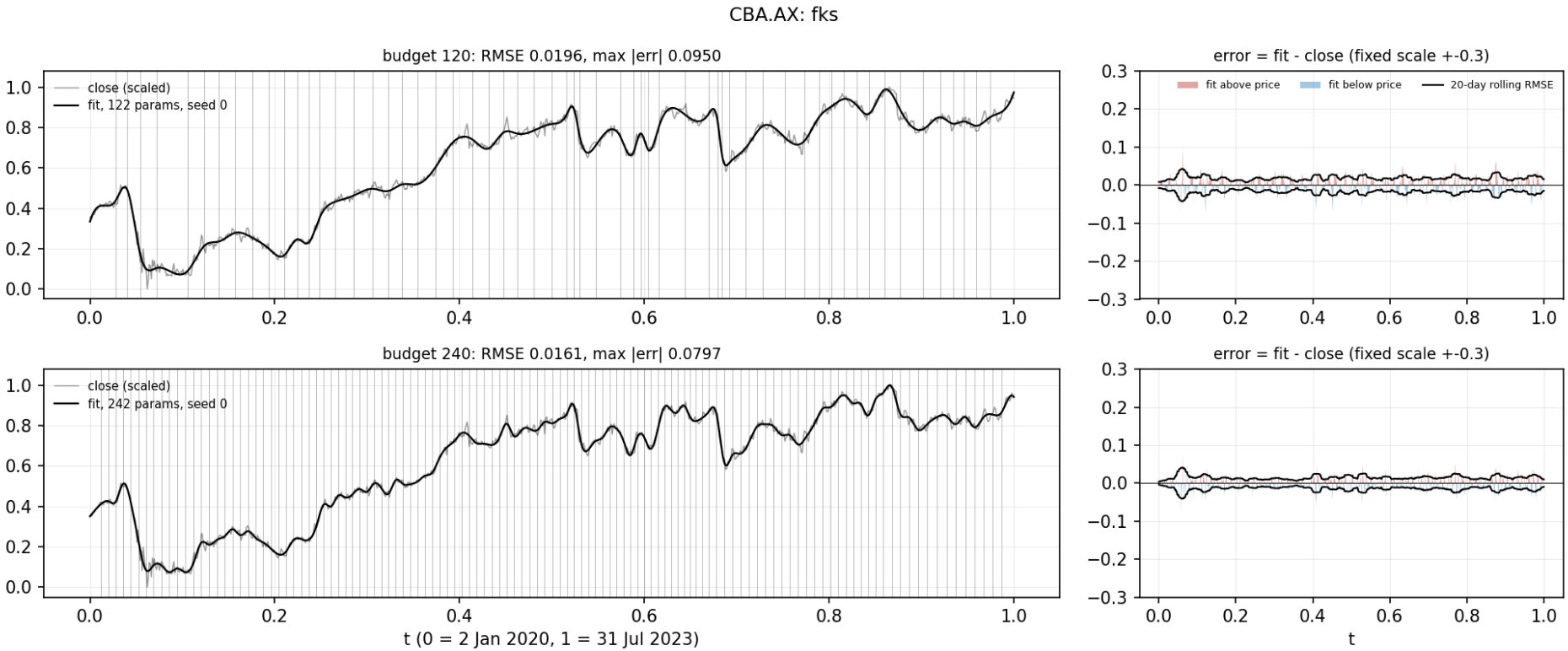
CBA.AX (Commonwealth Bank, benchmark series), Least-squares cubic spline, fixed uniform knots (budgets 120, 240)



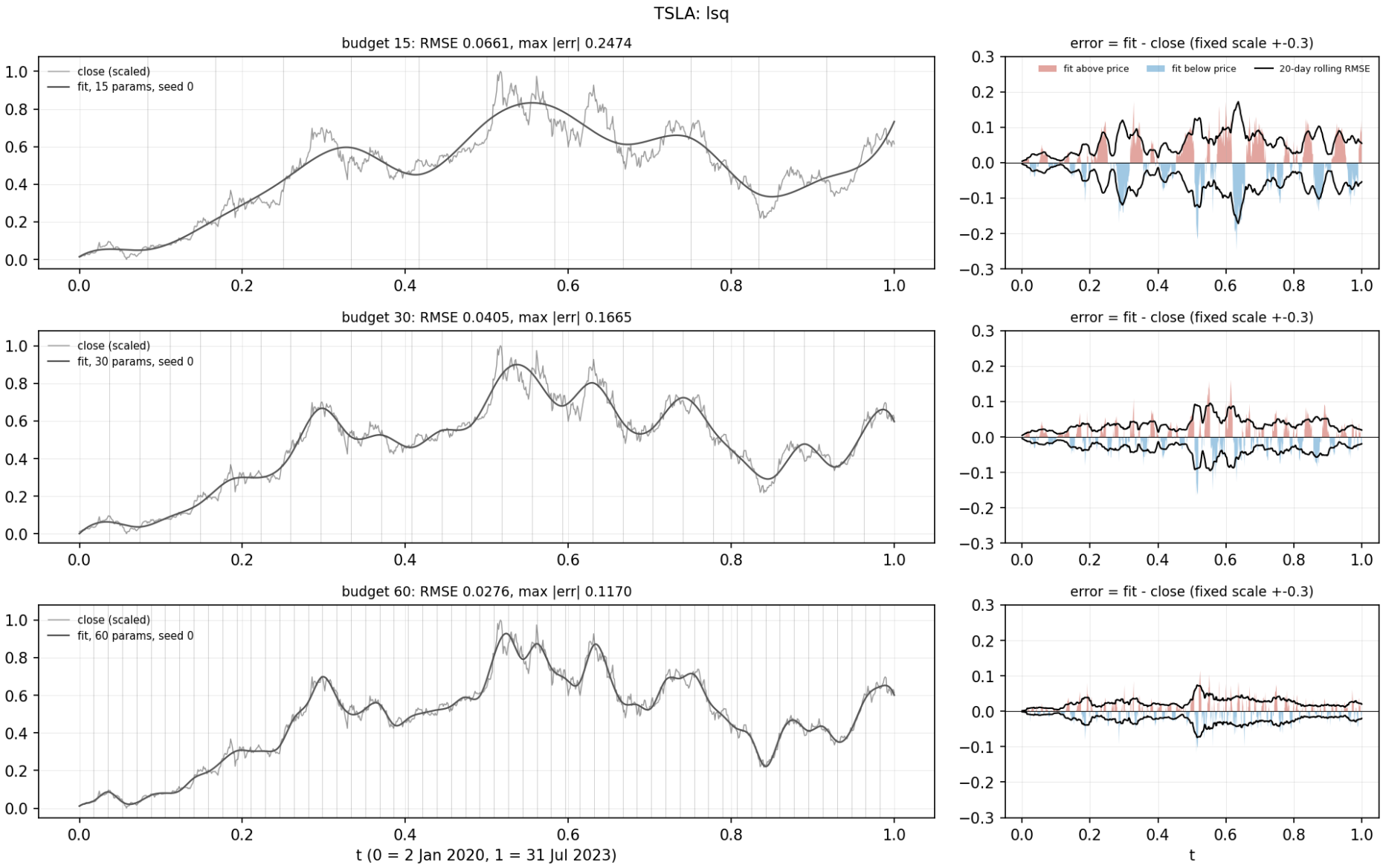
CBA.AX (Commonwealth Bank, benchmark series), Free-knot cubic spline, knot positions optimised (budgets 15, 30, 60)



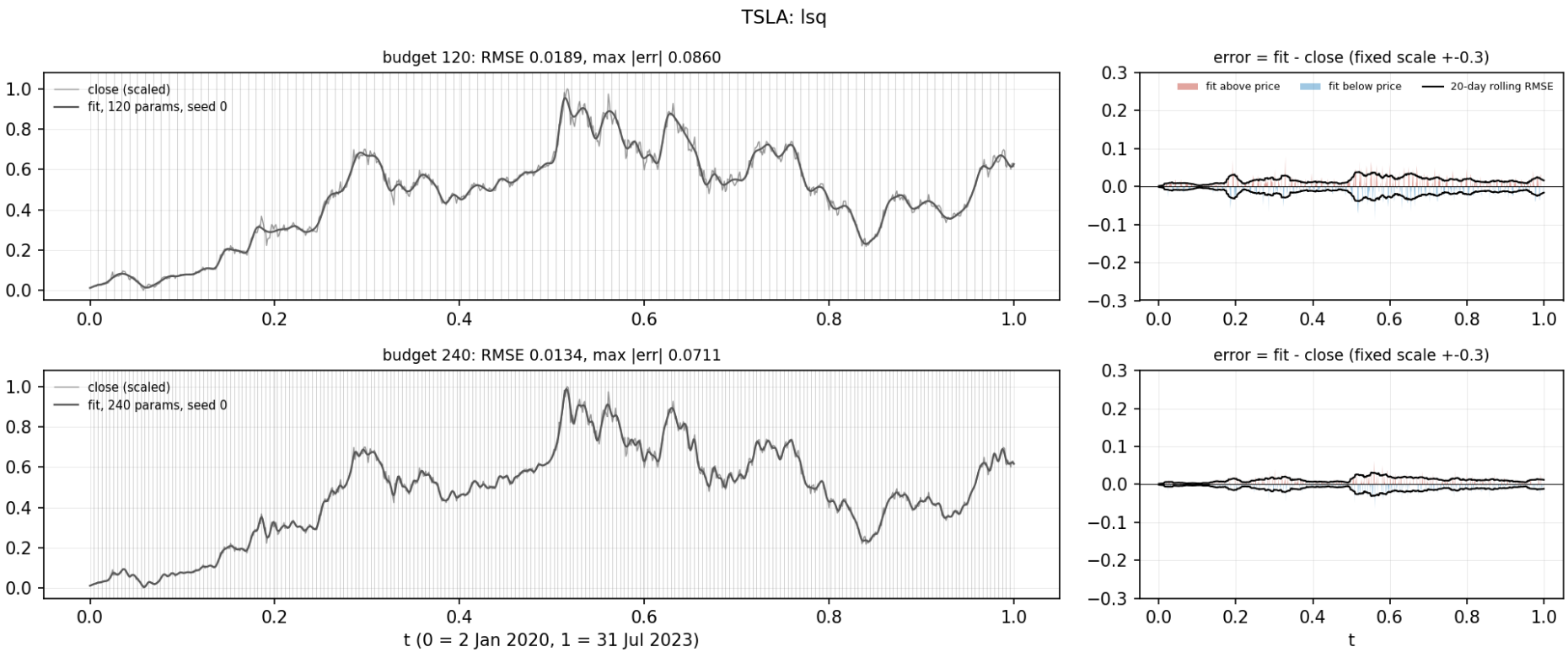
CBA.AX (Commonwealth Bank, benchmark series), Free-knot cubic spline, knot positions optimised (budgets 120, 240)



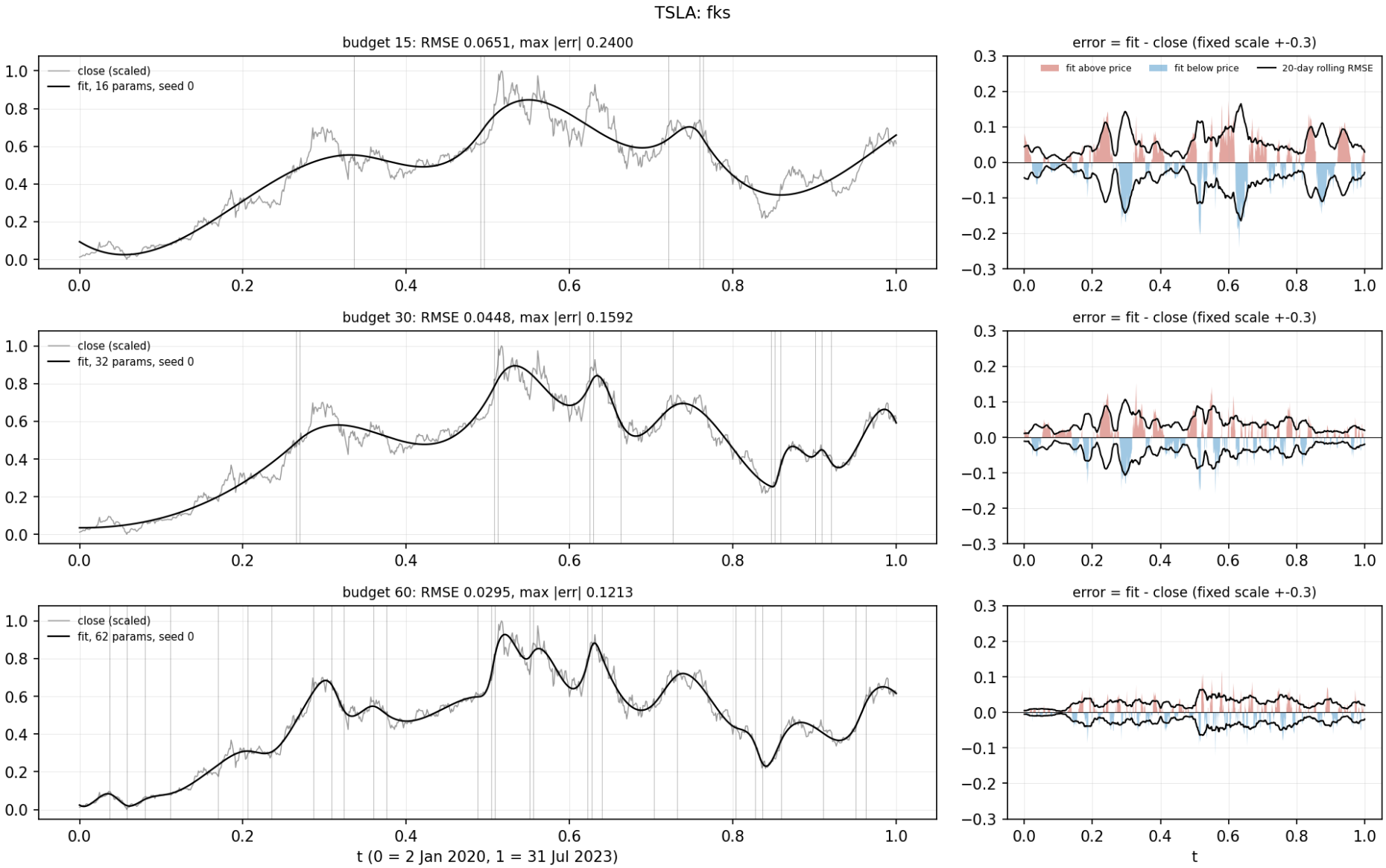
TSLA (Tesla, volatile), Least-squares cubic spline, fixed uniform knots (budgets 15, 30, 60)



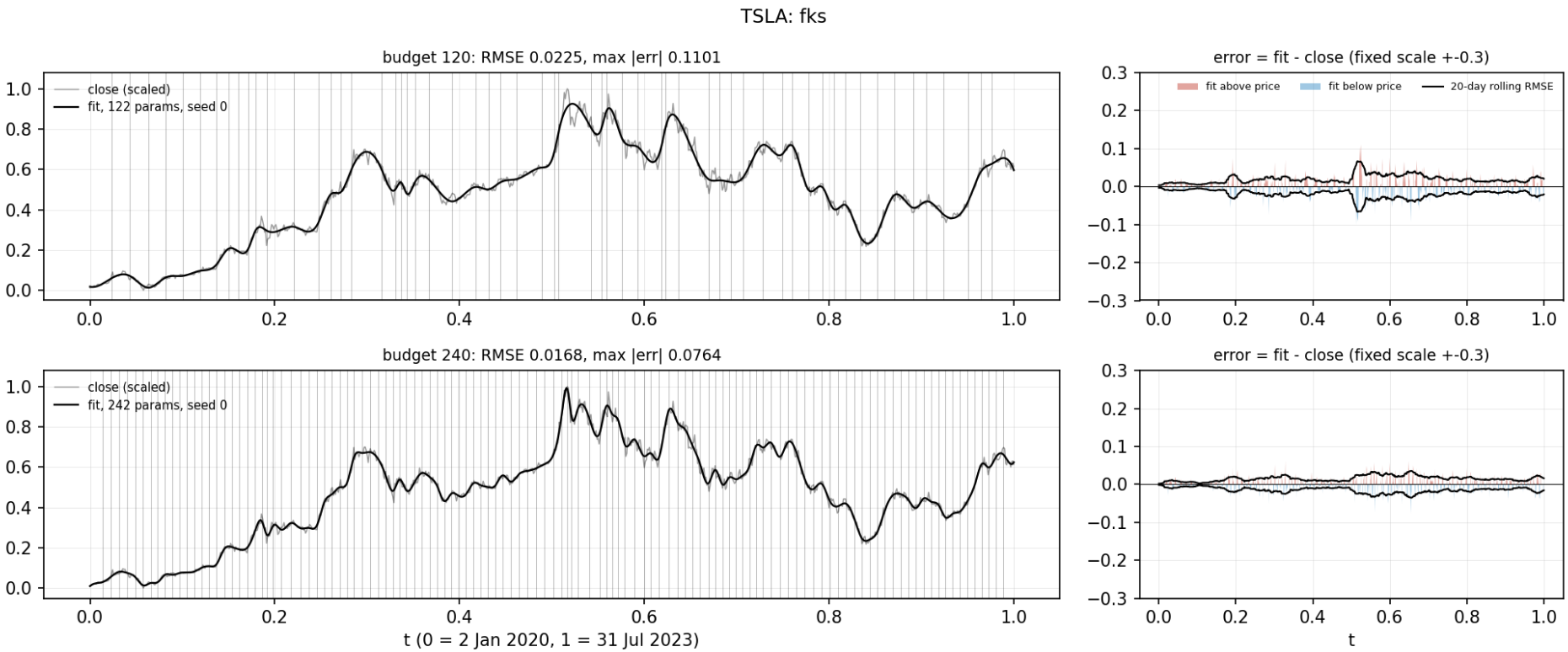
TSLA (Tesla, volatile), Least-squares cubic spline, fixed uniform knots (budgets 120, 240)



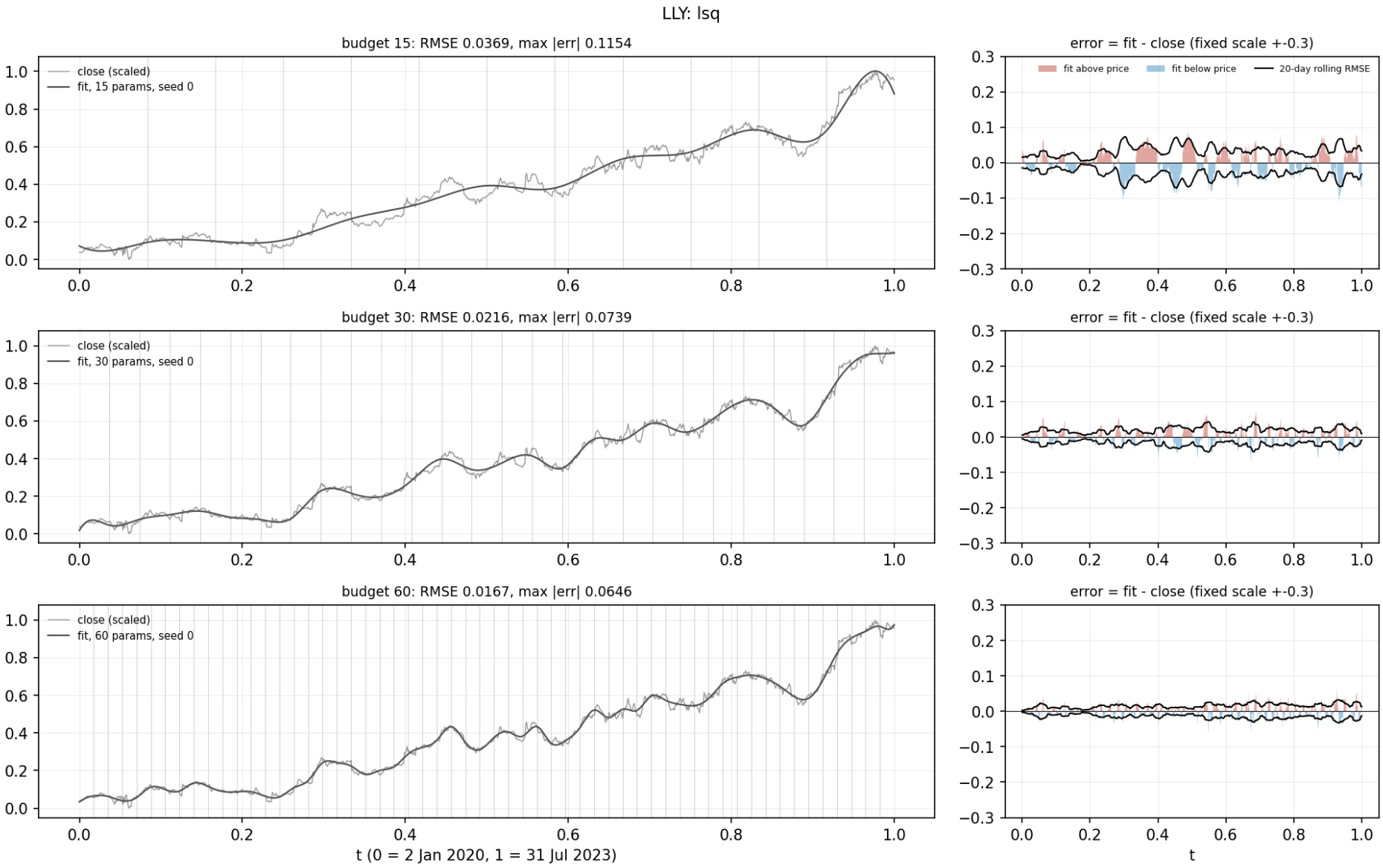
TSLA (Tesla, volatile), Free-knot cubic spline, knot positions optimised (budgets 15, 30, 60)



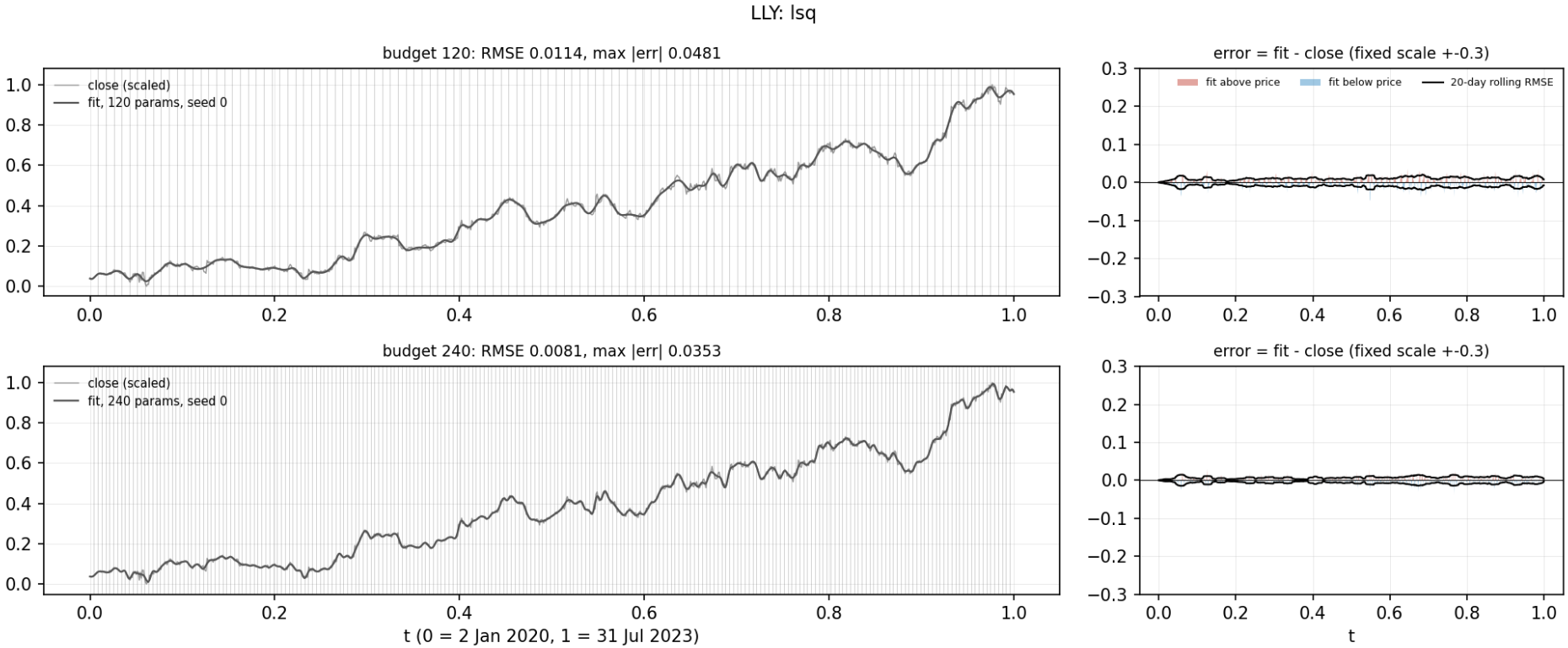
TSLA (Tesla, volatile), Free-knot cubic spline, knot positions optimised (budgets 120, 240)



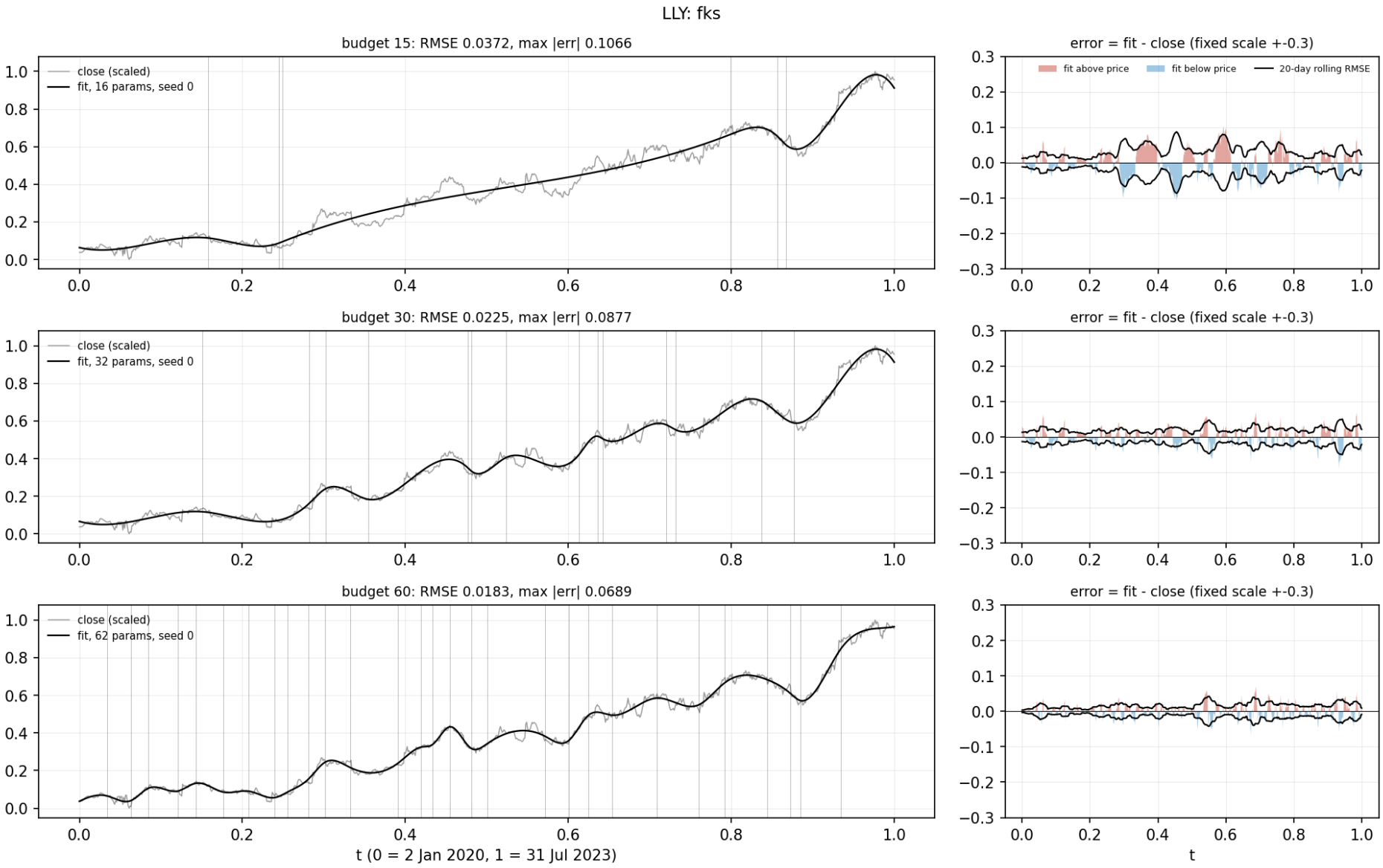
LLY (Eli Lilly, stable trend), Least-squares cubic spline, fixed uniform knots (budgets 15, 30, 60)



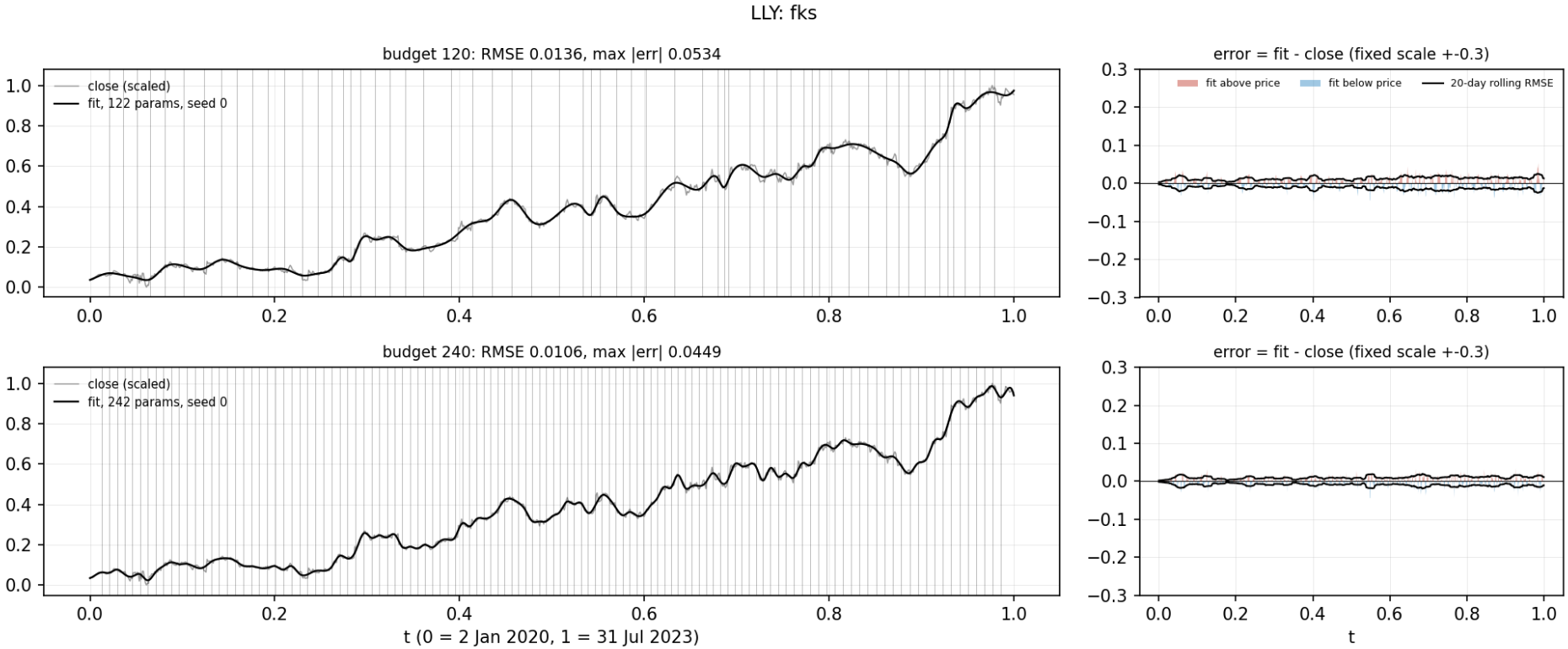
LLY (Eli Lilly, stable trend), Least-squares cubic spline, fixed uniform knots (budgets 120, 240)



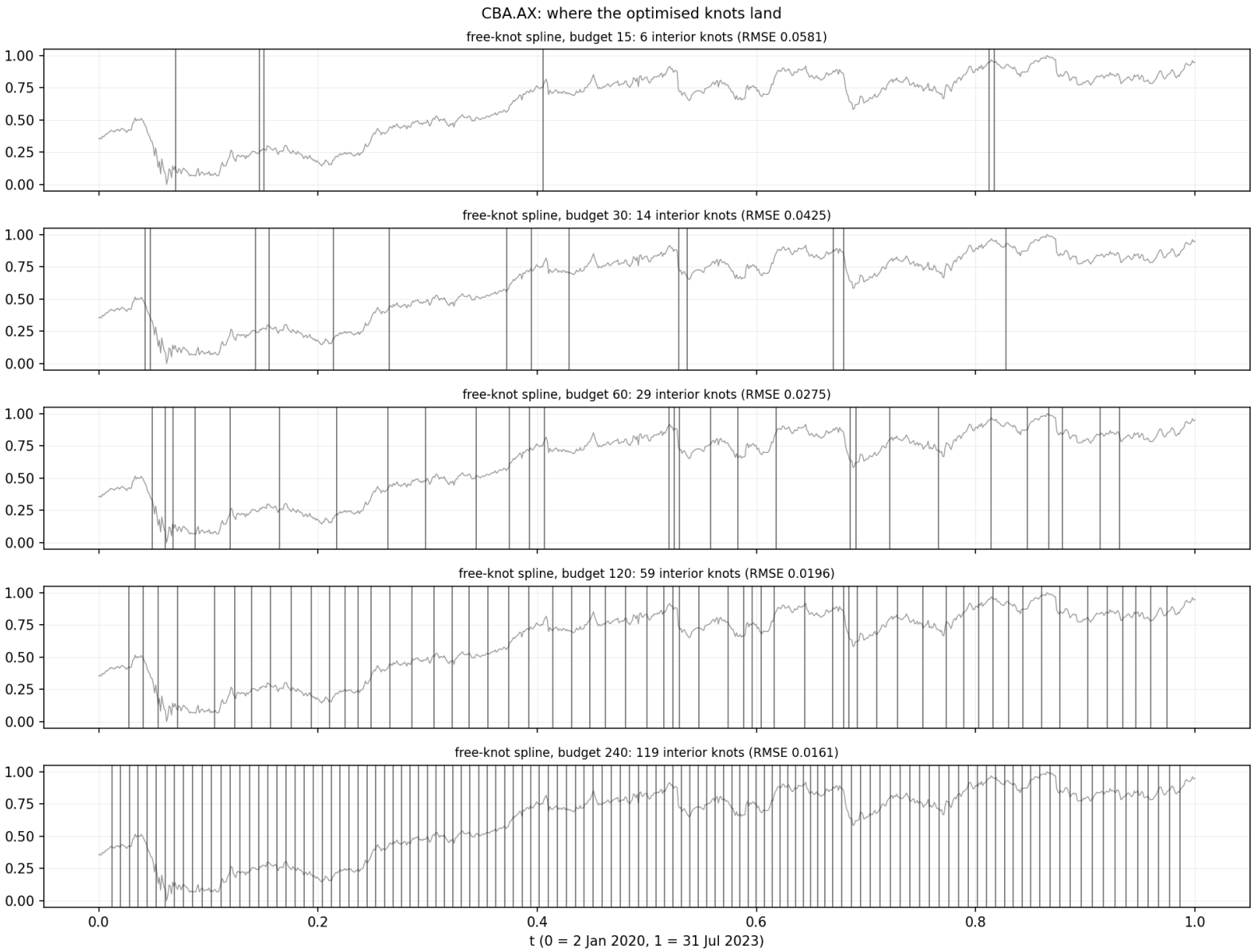
LLY (Eli Lilly, stable trend), Free-knot cubic spline, knot positions optimised (budgets 15, 30, 60)



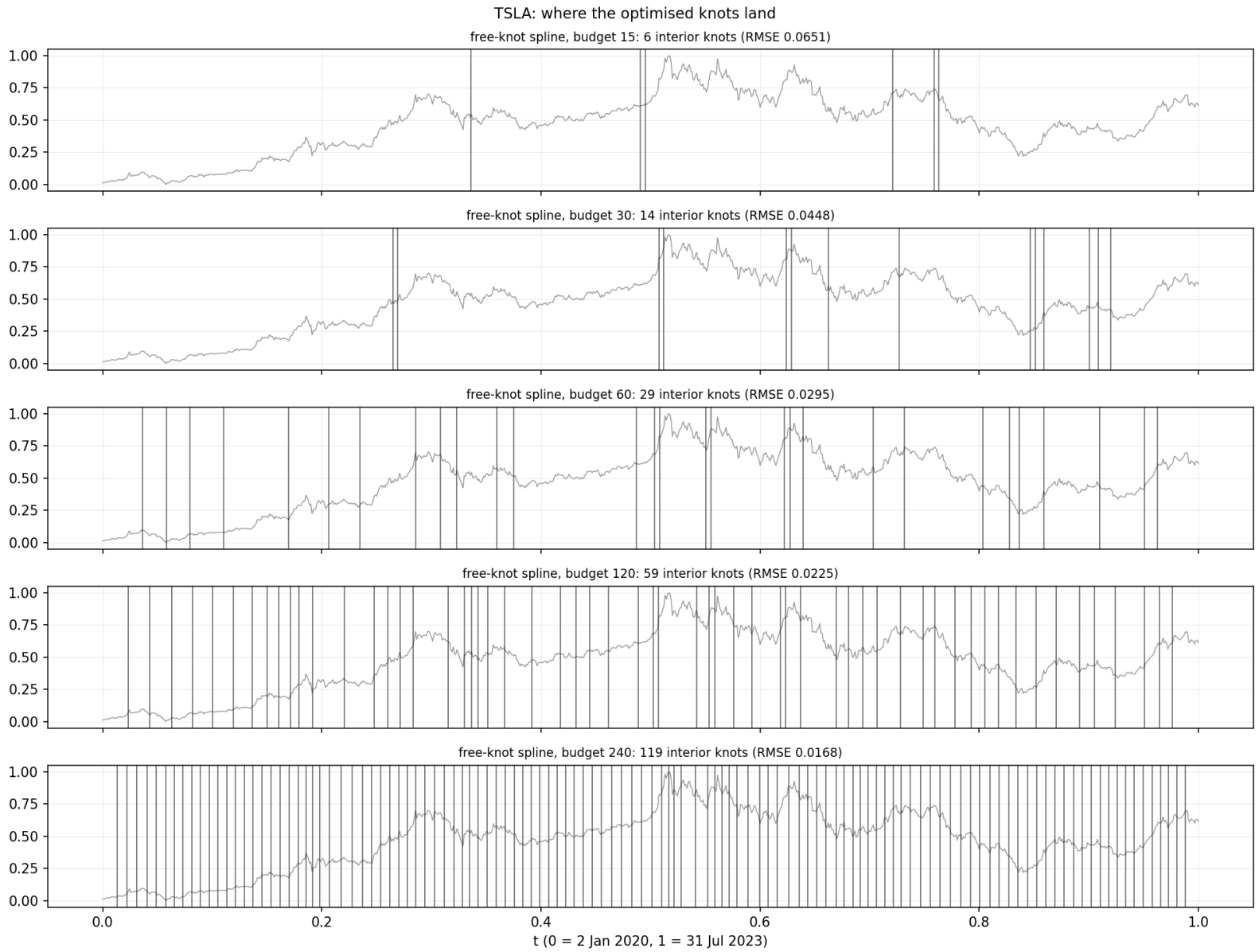
LLY (Eli Lilly, stable trend), Free-knot cubic spline, knot positions optimised (budgets 120, 240)



CBA.AX (Commonwealth Bank, benchmark series): optimised knot positions of the free-knot spline



TSLA (Tesla, volatile): optimised knot positions of the free-knot spline



LLY (Eli Lilly, stable trend): optimised knot positions of the free-knot spline

